

CRSS — Compliance Readiness Support System

Demo Report

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Question 1

If a company has a medical device and integrates an LLM into it for purposes outside of treating/diagnosing patients, would that still classify as a high-risk AI system? On the one hand it's integrated into a product covered by the legislation listed in Annex I, but on the other hand its purpose and application are not within the scope of the "medical device" definition.

Answer

ASSESSMENT STATUS — Provisional legal qualification assessment. This answer should be read as a case-specific compliance analysis, not as an automatic status determination.

1. Actor Status

The company that develops the AI system and integrates it into its own medical device is the provider of the AI system under **Article 3(3)** of the EU AI Act. It is also the manufacturer of the medical device under **Article 2** of the MDR.

2. Classification Route

The AI Act's high-risk classification follows **Article 6(1)** (Route I) for AI systems that are safety components of, or themselves, products covered by Union harmonisation legislation listed in **Annex I** (which includes the MDR) and that are required to undergo third-party conformity assessment under that legislation.

3. Trigger Event

The trigger is the placing on the market or putting into service of the AI system, either as part of the medical device or independently.

4. Legal Effects of Classification

4.1 AI Act High-Risk Classification

- **Article 6(1)(a)** requires the AI system to be a safety component of a product covered by **Annex I** (MDR).
- **Article 6(1)(b)** requires the product (the medical device) to be subject to third-party conformity assessment under the MDR.

The intended purpose of the AI system is decisive. The AI Act does not require the AI system itself to have a medical purpose; it only requires that the product (the medical device) is covered by the MDR and that the AI system acts as a safety component of that product.

- If the LLM is integrated into the medical device and its outputs influence the safety or performance of the device (e.g., by affecting clinical decision-making, device operation, or user interaction), it qualifies as a safety component under **Article 6(1)(a)**.
- The MDR classification of the medical device (e.g., Class IIa, IIb, or III) determines whether third-party conformity assessment is required. For Class IIa, IIb, or III devices, third-party conformity assessment is mandatory, satisfying **Article 6(1)(b)**.

If both conditions are met, the AI system is high-risk under the AI Act, regardless of whether its specific function is medical or non-medical.

4.2 MDR Classification

The medical device remains subject to the MDR, including its intended purpose as defined in **Article 2**. The integration of the LLM does not alter the device's classification under the

MDR, but it may require a significant change assessment under **Article 120(3)** if the LLM affects safety or performance.

5. Cross-Regulatory Chain

- The AI Act and MDR operate in parallel. The AI Act's high-risk classification does not depend on the LLM's medical purpose but on its role as a safety component of an MDR-covered product.
- The MDR governs the device's conformity assessment, while the AI Act imposes additional requirements on the AI system (e.g., risk management, technical documentation, transparency, and post-market monitoring) under Chapter III, Section 2.
- **Article 43(3)** of the AI Act ensures that the conformity assessment procedure for the AI system is integrated into the MDR conformity assessment, avoiding duplication.

6. Residual Uncertainty

- Safety Component Definition: The AI Act does not formally define "safety component." However, **Annex I** of the MDR (General Safety and Performance Requirements) establishes that any component affecting the device's safety or performance qualifies. If the LLM's outputs influence the device's operation, user interaction, or clinical outcomes, it is likely a safety component.
- Non-Medical Purpose: The LLM's non-medical purpose (e.g., administrative or patient communication) does not sever the nexus to the MDR-covered product. The AI Act's **Article 6(1)** focuses on the product's coverage under **Annex I**, not the AI system's specific function.
- Case-by-Case Assessment: The final determination depends on whether the LLM's integration affects the device's safety or performance. If it does, the AI system is high-risk under the AI Act.

Conclusion

The AI system is likely high-risk under the AI Act if: 1. It is integrated into a medical device covered by the MDR (**Annex I**). 2. The medical device is Class IIa, IIb, or III, requiring third-party conformity assessment (**Article 6(1)(b)**). 3. The LLM acts as a safety component by influencing the device's safety or performance (**Article 6(1)(a)**).

The LLM's non-medical purpose does not preclude high-risk classification under the AI Act. The decisive factor is its role as a safety component of an MDR-covered product. A case-by-case assessment is required to confirm whether the LLM's integration affects the device's safety or performance. If it does, the AI system is high-risk, and the company must comply with Chapter III, Section 2 of the AI Act, integrated into the MDR conformity assessment procedure under **Article 43(3)**.

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